

A Theory-Guided LLM Pedagogical Agent for STEM+C Scaffolding Without Dependence

Supplementary Materials

Overview

This supplement enumerates the prompt **features/inputs** and **outputs** for Copa’s multi-agent architecture. Raw prompt text is maintained separately in the repository¹; this document provides an interface-level specification suitable for replication.

Shared Task Context Features

All agents are prompted with a consistent task description. Table 1 enumerates the task-level constants and constraints that appear in the prompts, using the example 1-D XYZ truck task.

Table 1: Shared task context features (Truck Task).

Feature	Definition / Value
Task goal	Construct a computational model of a truck that accelerates from rest, cruises at a speed limit, and decelerates to stop at a stop sign.
Acceleration bounds	$\pm 4 \text{ m/s}^2$.
Speed limit	15 m/s.
Initial position	-60 (environment units).
Stop sign position	38.16.
Time step constant	<code>DeltaT = 0.1</code> .
State variables	<code>x_position</code> , <code>x_velocity</code> , <code>x_acceleration</code> .
Constants	<code>DeltaT</code> , <code>SpeedLimit</code> , <code>StopSignPosition</code> .
Execution model	Initialization logic under <code>When Green Flag Clicked</code> block; iterative updates under <code>Simulation Step</code> block.
Lookahead distance (ground truth)	28.125 (students must compute using kinematic equations; agents are instructed not to provide this value directly to students).
Testing tools	Students can run by clicking the Green Flag; Graph/Table tools may be used to visualize variable changes over time.
Runnable model structure	Two principal block groups: (1) [<code>When Green Flag Clicked</code>] for initialization; (2) [<code>Simulation Step</code>] for the loop.
Expert reference	One correct reference model is provided for logical equivalence checking (presented on the next page).

¹https://anonymous.4open.science/r/BJET26_Supplementary_Materials-5F5B

Expert reference model, translated from the block-based code on screen to textual form for the LLM by parsing the raw logs.

[When Green Flag Clicked]

Set DeltaT to 0.1

Set SpeedLimit to 15

Set x_position to -60

Set x_velocity to 0

Set x_acceleration to 4

[Simulation Step]

Change x_velocity by $(x_acceleration * \Delta T)$

Change x_position by $(x_velocity * \Delta T)$

If $x_velocity > SpeedLimit$ then

Set x_acceleration to 0

If $StopSignPosition - x_position < 28.125$ then

Set x_acceleration to -4

If $x_position > StopSignPosition$ and $x_velocity < 0$ then

Set x_velocity to 0

Stop Simulation

Agent 1: StrategyAgent (Evidence Module)

Purpose

Infer students' cognitive strategy from sequences of modeling actions in XYZ.

Action Vocabulary (Defined in the Prompt)

Table 2: Defined student action types used for strategy inference.

Action	Definition
BUILD	Add or connect runnable blocks affecting model execution.
ADJUST	Modify, remove, or reposition existing runnable blocks.
EXECUTE	Run the model using the Green Flag.
DRAFT	Add or modify disconnected (non-runnable) blocks.
VISUALIZE	Use graph or table tools to inspect model behavior.

Strategy Labels (Defined in the Prompt)

Table 3: StrategyAgent cognitive strategy labels. Thumbs up/down refer to good/bad strategies.

Strategy	Operational Definition
DEPTH_FIRST_ENACTING 🖱️	Extended BUILD/ADJUST sequences with no model execution.
TINKERING 👍	Frequent alternation between model edits and executions.
TOOL_USE 👍	High frequency of VISUALIZE actions during modeling.
RUN_REPEAT 🖱️	Three or more consecutive EXECUTE actions without edits.
DRAFTING 👍	Using disconnected blocks as placeholders, i.e., commenting code.

Output Schema

`{"summary" : string, "strategy" : one of Table 3}`

Agent 2: AssessmentAgent (Evidence Module)

Purpose

Infer the dyad’s learner state from model structure, strategy, and recent history.

Learner State Labels (Defined in the Prompt)

Table 4: Learner state definitions used by the AssessmentAgent.

Learner State	Definition
STARTING	No runnable blocks present in the student model.
PROGRESSING	Incomplete but logically correct model using effective strategies.
EXPLORING	Incomplete but correct model using suboptimal strategies.
DEBUGGING	Model contains at least one logical error.
STRUGGLING	Persistent debugging or lack of progress.
FINISHED	Model is complete and without error.

Output Schema

`{"summary" : string, "learner_state" : one of Table 4}`

Agent 3: KnowledgeAgent (Evidence Module)

Purpose

Identify the students' Zone of Proximal Development (ZPD), determine the next step the students must take in the environment, and select the domain knowledge to retrieve for scaffolding.

Input

Student's computational model (i.e., their code) and the expert model presented earlier.

Output Schema

```
{"summary": string, "recommended_domain_knowledge": string}
```

Agent 4: DialogueAgent (Decision and Feedback Modules)

Purpose

The DialogueAgent is responsible for generating adaptive, student-facing feedback. It operates in two explicit stages: **(1) dialogue state inference** based on the current interaction context, and **(2) feedback generation** conditioned on the inferred state, learner state, and pedagogical policy constraints.

Dialogue State Labels (Defined in the Prompt)

Table 5: Dialogue state definitions.

Dialogue State	Definition
OFF_TASK	Utterance unrelated to the task.
DISENGAGED	Minimal or apathetic response.
ENACTING	Student describing ongoing modeling actions.
FRUSTRATION	Expression of confusion or difficulty.
INFORMATION_SEEKING	Direct question about the XYZ task/domain concepts.
ACKNOWLEDGING	Student acknowledges prior feedback.
SUGGESTING	Student proposes a next action.
UNDERSTANDING_RESPONSE	Evidence of conceptual understanding.
NOT_UNDERSTANDING_RESPONSE	Evidence of misunderstanding.

Dialogue Policy Labels (Defined in the Prompt)

Table 6: Dialogue policy definitions.

Policy	Definition
PROBE_UNDERSTANDING	Ask a question to elicit reasoning.
SET_GOAL	Help students articulate a short-term objective.
SUGGEST_ACTION	Propose a concrete next modeling step.
PROVIDE_DOMAIN_KNOWLEDGE	Share relevant conceptual knowledge.
PROMOTE_SELF_EFFICACY	Encourage confidence and persistence.
STRATEGY_RECOMMENDATION	Suggest a more productive learning strategy.
SUPPORT_ENGAGEMENT	Re-engage off-task or disengaged students.
AWAIT_ACTION	Pause feedback while students act.
SEEK_HELP	Recommend external help after repeated failure.
END_GAME	Acknowledge task completion and conclude.
GET_CLARIFICATION	Request clarification (policy only; not a state).

Response Constraints

- Responses must be ≤ 100 characters.
- Responses must maintain a collaborative peer persona and adhere to social cognitive and social constructivist principles (defined in the prompt).
- Hidden inputs (states, mastery, retrieved knowledge) must not be disclosed.

Output Schema

The DialogueAgent must return a JSON object with the following required fields:

```
{
  "dialogue_state_summary": string,
  "dialogue_state": dialogue state label,
  "dialogue_policy_summary": string,
  "dialogue_policy": one or two dialogue policy labels,
  "response": string
}
```